University of Glasgow

James Watt School of Engineering

Degrees of MEng, BEng, MSc and BSc in Engineering

ENG1064 Microelectronic Systems 1

Monday, 13 May 2024, Release time:14:00-15:30

Exam duration: 1.5 hours to complete exam plus 30 mins to upload submission

Attempt ALL FOUR questions

TOTAL MARKS AVAILABLE
50

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

A calculator may be used. Show intermediate steps in calculations.

OVER

Q1 – Embedded Systems

Most of the electronic devices around us are based on microcontroller electronics and are built to continuously perform a particular task.

a) Briefly explain what an embedded system is. Detail the main functionalities of such a system and its differences from a personal computer. [3]

b) Below, Figure Q1(a), shows an example of a vending machine. This is an embedded system with many of its functionalities based on microcontroller electronics.
   i. Detail the embedded functionalities of a vending machine, i.e. what are the inputs, outputs and what information is being processed by the microcontroller. (Use a maximum of 200 words) [3]
   ii. In modern vending machines, instead of inserting coins, you can directly pay with your credit card. Describe what embedded functionalities can make this possible. [3]
   iii. You are an embedded systems developer working for a big vending machine manufacturer. Your company is developing the next generation of smart vending machines and are looking for unique functionalities to incorporate in the next generation vending machines.

   Develop and compose a new, smart feature for a vending machine. This can be an imaginative feature. Make all the necessary links to embedded functionalities, i.e. what are the input/output signals, and information is being processed. (200 words max) [2]

Figure Q1a: A vending machine

Continued overleaf

Q2 – Number conversion

Digital electronics is based on the binary number system, where 0 and 1 are the only numbers used, and this is a number system with base-2 (2^N). Imagine that you are sent to a parallel universe where instead of a binary number system people used a new number system with their computers. This is called **quadinary** system and it is based on **quads** instead of our usual bits. The **quadinary** is a **base 4** (4^N) number system and quads can have values of [0,1,2,3]. By extending your knowledge on the binary system, please answer the questions below.

a) For an 8-quad system, what is the maximum decimal number we can create? [2]

b) Convert the quadinary number 11000010 to decimal. [2]

c) Verify that the quadinary number 13030210 corresponds to 29476 in the decimal number system. [3]

d) Convert the quadinary number 30223332 to the hexadecimal system (0,1,2,3,4,5,6,7,8,9, A,B,C,D,E,F)
(Note: Draw from your knowledge on converting from binary to hexadecimal, but instead of doing it in blocks of 4 you should ***do it in blocks of 2****).* [4]

e) In this imaginary world with the **quadinary** number system, a microcontroller has address and data buses made out of 6 parallel wires. Assuming each wire can carry one quad, what is the maximum number of memory locations it can have? [1]

A 7-quad Analogue-to-Digital Converter (ADC) in this parallel universe is connected to a reference voltage of 1 V.

f) Calculate the resolution in mV (milli-volts) [1]
g) Calculate the range in V (volts) (give accuracy of 5 decimal points). [1]

Continued overleaf

Q3 – MCU inputs and Outputs

The Nucleo microcontroller has several pins that can be configured as Digital Outputs. Each of these pins can provide a DC voltage of 3.3 Volts.

a) Summarise how you would connect a switch to the microcontroller. [2]
b) Describe in detail, using appropriate diagrams, how you would connect a red LED to one of the MCU pins. The LED operates with a voltage of **1.75 Volts** and **2 mA** current. Label all the components and calculate their values if necessary. [3]
c) Explain if it is possible to connect a second red LED to the same port. Illustrate with a new schematic how you would connect it and label the required components accordingly. [5]
d) You would like to connect a motor to one of the PWM (Pulse Width Modulation) outputs of your microcontroller. For normal operation, the motor requires a voltage of 24 Volts. Explain how you would connect the motor, using appropriate diagrams. [5]

Continued overleaf

Q4 – SPI communication protocol

We would like to establish a serial communication protocol, using our Microcontroller (MCU) as a Master and 6 other sensors designated as Slaves. The serial protocol should be based on SPI (Serial Peripheral Interface).

a) How many connections in total will we need to connect to our microcontroller? Briefly justify your answer. [2]

b) Data will be sent to the slaves in the following format, Slave-1: 4 bits, Slave-2: 8 bits, Slave-3: 12 bits, Slave-4: 16 bits, Slave-5: 1 bit and Slave-6: 4 bits. All the data transmission has to be completed within 100 μs (micro-seconds). Assuming there are no other 'time overheads', what would be the minimum clock frequency you can use? [2]

c) You would like to send to Slave-2 the hex-word 0xB3. Assuming you use the maximum frequency (80 MHz), plot the signal which is sent to Slave-2 as a function of time. Label appropriately your plot. [6]

End of question paper